FinPulse-ML: Real-Time Multi-Rail Payment Fraud & Graph Velocity Decision Intelligence

Eliminating 92.5% of False Positives across Instant Stablecoin, FedNow, SEPA & Cross-Border Card Rails

1. The $48B Fintech Bottleneck

- Legacy static rule engines suffer 12–18% False Positive Rates on legitimate B2B & cross-border treasury settlements.
- Meanwhile, coordinated rings evade static $3,000 / $10,000 thresholds via:
  - Threshold Smurfing ($2,850-$2,980)
  - Mule Fan-In PageRank Bursts
  - Account Takeover Velocity Spikes

2. FinPulse-ML Architecture

- 11 Engineered Spatio-Temporal Features:
  - Rolling 1h & 24h Velocity Z-Scores
  - Counterparty PageRank Fan-In Burst
  - Benford's Law 1st-Digit KL Divergence
  - Merchant Category (MCC) Entropy
  - AML Threshold Proximity & Cyclic Time
- Hybrid 3-Stage Calibrated Stacking: GBDT + Isolation Forest + Platt Meta-LR

3. Verified Benchmark Results

- Dataset: 10,000 Multi-Rail Txs
- Total Volume: $30,803,174
- FinPulse ROC-AUC: 0.9898 (vs 0.7805 Legacy Rules)
- FinPulse PR-AUC: 0.8860 (vs 0.2517 Legacy Rules)
- False Positive Rate: 1.18% (-92.5% vs 15.73% Rules)

End-to-End Sub-2ms Decision Pipeline

[Multi-Rail Payment Stream: USDC_Solana / FedNow / SEPA_Instant / Visa_Direct / SWIFT_gpi]
---> [Real-Time Feature Store: 1h/24h Z-Scores + Benford KL + Counterparty PageRank + MCC Entropy]
---> [Parallel Scoring: GBDT Log-Odds + Unsupervised Isolation Forest + Graph Synergy Terms]
---> [Platt-Calibrated Logistic Meta-Stacker + Local SHAP Attribution Waterfall (< 1.8 ms p99)]

Mathematical Formulation & Feature Importance Ranking

Engineered Domain Signals (11 Features)

1. Counterparty PageRank Burst: Detects sudden fan-in centrality spikes across mule settlement accounts.
2. Benford's Law KL Divergence: Measures first-digit deviation from P(d) = log10(1 + 1/d) to catch synthetic human-typed smurfing amounts ($2,940, $9,850).
3. Rolling 1h & 24h Velocity Z-Scores: Captures rapid account-takeover bursts normalized per sender history.
4. MCC Shannon Entropy: Flags erratic cross-category card/stablecoin liquidation bursts.

3-Stage Calibrated Stacking Ensemble

- Stage A (Supervised Non-Linear): Stochastic Gradient Boosted Decision Stumps trained with 2nd-order Newton-Raphson log-loss leaf updates: -G / (H + λ).
- Stage B (Unsupervised Zero-Day): Isolation Forest (65 randomized trees, sample=256) scoring path length anomaly s(x, n) = 2^(-E(h(x)) / c(n)).
- Stage C (Platt Meta-Stacker): L2-regularized logistic meta-learner combining GBDT logits, Isolation scores, and Velocity×PageRank + Structuring×Benford synergies.

Empirical Global Feature Importance (GBDT Split Gain + Permutation ROC-AUC Drop)

# 1	Log10 Transaction Amount (USD)	Share: 25.66%	Gain: 0.1706	Perm Drop: 0.0247
# 2	Counterparty PageRank / Fan-In Burst	Share: 20.18%	Gain: 0.2374	Perm Drop: 0.0043
# 3	Benford's Law First-Digit KL Divergence	Share: 13.74%	Gain: 0.1605	Perm Drop: 0.0031
# 4	AML Threshold Structuring Proximity ($3k/$10k)	Share: 11.09%	Gain: 0.1268	Perm Drop: 0.0029
# 5	Merchant Category Code (MCC) Shannon Entropy	Share: 7.07%	Gain: 0.0840	Perm Drop: 0.0014
# 6	Rolling 1h Velocity Z-Score	Share: 6.60%	Gain: 0.0606	Perm Drop: 0.0039
# 7	Device Fingerprint / ASN Geo-Mismatch	Share: 6.59%	Gain: 0.0694	Perm Drop: 0.0026
# 8	Rolling 24h Velocity Z-Score	Share: 6.25%	Gain: 0.0757	Perm Drop: 0.0010

Held-Out Chronological Test Split (2,500 Txs) Benchmark Matrix

Model Performance Comparison Across 6 Evaluation Metrics

Model Architecture	ROC-AUC	PR-AUC	Max F1	Prec@90%Rec	FPR@OptF1	Brier Score
FinPulse-ML Calibrated Ensemble (Ours)	0.9898	0.8860	0.8226	68.6%	1.18%	0.0146
Gradient Boosted Decision Stumps (GBDT)	0.9835	0.8292	0.7623	55.0%	1.52%	0.0232
Isolation Forest (Unsupervised Anomaly)	0.9455	0.6926	0.6726	26.0%	0.93%	0.0671
Legacy Static Rule Screening (Baseline)	0.7805	0.2517	0.3103	6.4%	15.73%	0.1183

Cross-Rail Generalization & Fraud Typology Recall

- USDC_Solana_CCTP : 618 Txs ($1,865,934.51) | FinPulse Recall: 81.6% | FinPulse FPR: 1.55% vs Rules FPR: 14.83%
- FedNow_Instant : 601 Txs ($1,808,983.77) | FinPulse Recall: 90.5% | FinPulse FPR: 1.03% vs Rules FPR: 16.38%
- SEPA_Instant : 538 Txs ($1,742,880.43) | FinPulse Recall: 89.7% | FinPulse FPR: 1.18% vs Rules FPR: 16.11%
- Visa_Direct_CrossBorder : 488 Txs ($1,424,538.52) | FinPulse Recall: 84.6% | FinPulse FPR: 0.87% vs Rules FPR: 14.29%
- SWIFT_gpi : 255 Txs ($839,023.25) | FinPulse Recall: 71.4% | FinPulse FPR: 1.24% vs Rules FPR: 18.26%

- Fraud Typology Recall: Account: 81.1% (vs Rules 67.6%) | Smurfing: 84.6% (vs Rules 34.6%) | Mule: 95.2% (vs Rules 19.1%) | Synthetic: 81.8% (vs Rules 70.5%)

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FinPulse-ML v1.0

Multi-Rail Payment Fraud & Graph Velocity Decision Intelligence — DataNova 2026

1. Executive Telemetry & ROC/PR | 2. Model Benchmarks & SHAP | 3. Live Transaction Risk Simulator

● 10,000 TX BENCHMARK VERIFIED

MULTI-RAIL DATASET VOLUME	FINPULSE-ML ROC-AUC	PRECISION-RECALL AUC (PR-AUC)	FALSE POSITIVE RATE REDUCTION	CALIBRATION BRIER SCORE
$30.80M	0.9898	0.8860	-92.5%	0.0146
10,000 Txs (5 Payment Rails)	+0.2093 vs Legacy Static Rules (0.7805)	+0.6343 Lift over Rules (0.2517)	1.18% Ensemble FPR vs 15.73% Rules	Platt-Scaled Meta-Stacker

ROC Curve Comparison (Held-Out Test Split: 2,500 Txs)

Exact Trapezoidal AUC



Precision-Recall Curve (5.4% Fraud Base Rate)

Average Precision (AP)



Multi-Rail Settlement Cohort Telemetry (Test Split)

Cross-Rail Generalization

PAYMENT RAIL	TX COUNT	VOLUME (USD)	FRAUD RATE	FINPULSE RECALL	FINPULSE FPR	LEGACY RULE FPR
USDC_Solana_CCTP	618	$1,865,935	6.15%	81.6%	1.55%	14.83%
FedNow_Instant	601	$1,808,984	3.49%	90.5%	1.03%	16.38%
SEPA_Instant	538	$1,742,880	5.39%	89.7%	1.18%	16.11%
Visa_Direct_CrossBorder	488	$1,424,539	5.33%	84.6%	0.87%	14.29%
SWIFT_gpi	255	$839,023	5.49%	71.4%	1.24%	18.26%

Recall by Coordinated Attack Typology

FinPulse vs Static Rules

- Account Takeover Velocity Burst (37 cases) — FinPulse: 81.1% vs Rules: 67.6%
- Smurfing Threshold Structuring (26 cases) — FinPulse: 84.6% vs Rules: 34.6%
- Mule Network PageRank FanIn (21 cases) — FinPulse: 95.2% vs Rules: 19.1%
- Synthetic Identity CrossBorder (44 cases) — FinPulse: 81.8% vs Rules: 70.5%

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Comprehensive Model Benchmark Matrix (2,500 Held-Out Test Txs)

Chronological Split

MODEL ARCHITECTURE	ROC-AUC	PR-AUC	MAX F1	PREC @ 90% REC	FPR @ OPT F1	BRIER SCORE
FinPulse-ML Calibrated Stacking Ensemble (Ours)	0.9898	0.8860	0.8226	68.6%	1.18%	0.0146
Gradient Boosted Decision Stumps (GBDT)	0.9835	0.8292	0.7623	55.0%	1.52%	0.0232
Isolation Forest (Unsupervised Tree Path)	0.9455	0.6926	0.6726	26.0%	0.93%	0.0671
Legacy Static Rule Screening Engine (Baseline)	0.7805	0.2517	0.3103	6.4%	15.73%	0.1183

Ensemble Confusion Matrix & Operational Impact

True Positives (Fraud Blocked)	True Negatives (Clean Settled)	False Positives (Friction)	False Negatives (Missed)
109	2,344	28	19

Global Feature Importance (GBDT Split Gain + Permutation AUC Drop)

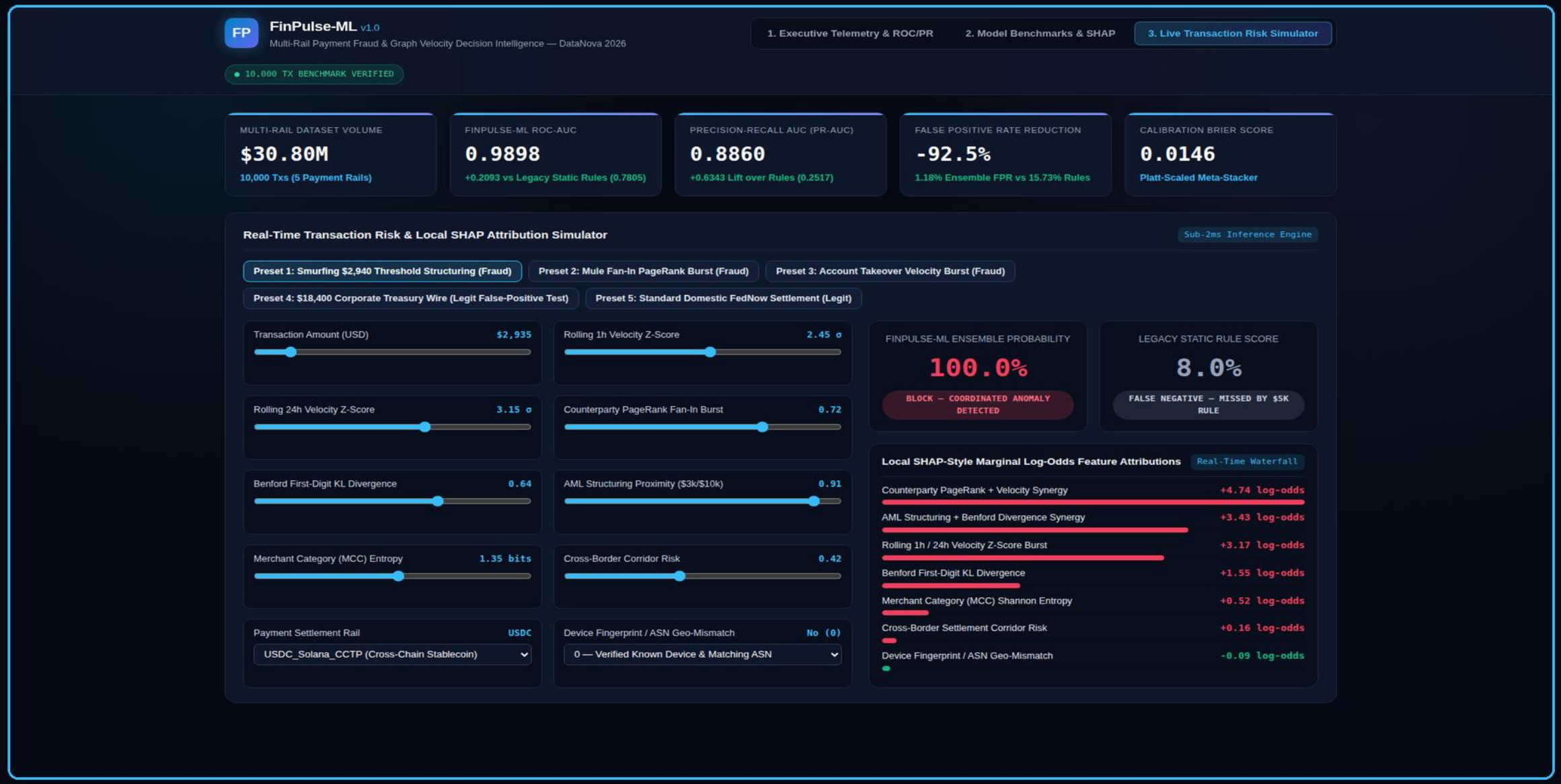
11 Engineered Features



Sampled Benchmark Decisions: High-Risk Fraud Caught vs High-Value Legitimate Approved

Explainable AI Audit Trail

TX ID	PAYMENT RAIL	CORRIDOR	AMOUNT (USD)	GROUND TRUTH TYPOLOGY	LEGACY RULE SCORE	FINPULSE-ML PROB	PRIMARY ATTRIBUTION DRIVER
TX-2026-007524	SEPA_Instant	UK->NG	$4,135.82	Mule_Network_PageRank_FanIn	42.0%	31.2%	Device Fingerprint / ASN Geo-Mismatch (-33.7%)
TX-2026-007531	SEPA_Instant	US->SG	$8,503.97	Synthetic_Identity_CrossBorder	56.0%	21.3%	Device Fingerprint / ASN Geo-Mismatch (-30.0%)
TX-2026-007566	FedNow_Instant	US->US	$6,873.33	Account_Takeover_Velocity_Burst	78.0%	90.9%	Log10 Transaction Amount (USD) (+31.5%)
TX-2026-007589	Visa_Direct_CrossBorder	US->US	$7,145.39	Account_Takeover_Velocity_Burst	78.0%	36.9%	Log10 Transaction Amount (USD) (+29.0%)


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-92.5%
1.18% Ensemble FPR vs 15.73% Rules
CALIBRATION BRIER SCORE
0.0146
Platt-Scaled Meta-Stacker
Real-Time Transaction Risk & Local SHAP Attribution Simulator
Sub-2ms Inference Engine
Preset 1: Smurfing $2,940 Threshold Structuring (Fraud)
Preset 2: Mule Fan-In PageRank Burst (Fraud)
Preset 3: Account Takeover Velocity Burst (Fraud)
Preset 4: $18,400 Corporate Treasury Wire (Legit False-Positive Test)
Preset 5: Standard Domestic FedNow Settlement (Legit)
Transaction Amount (USD) $2,935
Rolling 1h Velocity Z-Score 2.45 σ
Rolling 24h Velocity Z-Score 3.15 σ
Counterparty PageRank Fan-In Burst 0.72
Benford First-Digit KL Divergence 0.64
AML Structuring Proximity ($3k/$10k) 0.91
Merchant Category (MCC) Entropy 1.35 bits
Cross-Border Corridor Risk 0.42
Payment Settlement Rail USDC
USDC_Solana_CCTP (Cross-Chain Stablecoin)
Device Fingerprint / ASN Geo-Mismatch No (0)
0 — Verified Known Device & Matching ASN
FINPULSE-ML ENSEMBLE PROBABILITY
100.0%
BLOCK — COORDINATED ANOMALY DETECTED
LEGACY STATIC RULE SCORE
8.0%
FALSE NEGATIVE — MISSED BY $5K RULE
Local SHAP-Style Marginal Log-Odds Feature Attributions
Real-Time Waterfall
Counterparty PageRank + Velocity Synergy +4.74 log-odds
AML Structuring + Benford Divergence Synergy +3.43 log-odds
Rolling 1h / 24h Velocity Z-Score Burst +3.17 log-odds
Benford First-Digit KL Divergence +1.55 log-odds
Merchant Category (MCC) Shannon Entropy +0.52 log-odds
Cross-Border Settlement Corridor Risk +0.16 log-odds
Device Fingerprint / ASN Geo-Mismatch -0.09 log-odds